

SHUAIHANG CHEN

Robotics PhD Candidate · VLA Post-Training · Robot Reward Modeling

s-chensh24@bza.edu.cn · [Website](#) · [GitHub](#) · [Google Scholar](#)

SUMMARY

Robotics PhD Candidate working on Vision-Language-Action post-training and robot reward modeling. My research focuses on learning from mixed-quality robot trajectories, reinforcement-learning-based sim-real co-training, and scalable progress and quality signals for real-world manipulation.

RESEARCH

Robot Reward Modeling for VLA Post-Training

2026 – Present

Ongoing Research

- Investigating general-purpose reward and value models for evaluating Vision-Language-Action policies on real-world manipulation tasks.
- Building scalable training and evaluation pipelines for VLA post-training using robot trajectories with diverse task outcomes.

RL-Based Sim-Real Co-Training for VLA Models

2025 – 2026

Co-First Author [Project Page](#) [Paper](#)

- Led end-to-end OpenVLA post-training and evaluation across four real-world manipulation tasks, improving average success rate from 40% to 64% over the behavior-cloning-based sim-real co-training baseline.
- Implemented a two-stage training recipe consisting of supervised fine-tuning on mixed real and simulated demonstrations, followed by reinforcement learning in simulation with real-data regularization.
- Built simulation environments for two of the four tasks, supporting policy training, evaluation, and sim-to-real transfer under substantial visual and physical mismatch.
- Conducted data-scaling studies showing that RL co-training with 50 real trajectories outperformed pure supervised fine-tuning using 200 real trajectories in the evaluated setting.

STEAM: Self-Supervised Temporal Ensemble Advantage Modeling for Real-World Robot Learning

2026

Co-Author [Paper](#)

- Co-authored a collaborative study on self-supervised frame-level advantage estimation from expert temporal structure for identifying useful transitions in mixed-quality robot data and improving policy learning.

EDUCATION

Zhongguancun Academy & Harbin Institute of Technology

2024 – Present

Ph.D. Student in Robotics, Joint Program

Advisor: [Chao Yu](#)

Harbin Institute of Technology

2020 – 2024

B.Eng. in Software Engineering

EXPERTISE

Research Robot Reward and Advantage Modeling, VLA Post-Training, Mixed-Quality Trajectory Learning, Sim-Real Reinforcement Learning

Systems OpenVLA and π -family VLA training and evaluation, RLinf-based training pipelines, ManiSkill simulation and task construction, real-to-sim trajectory replay